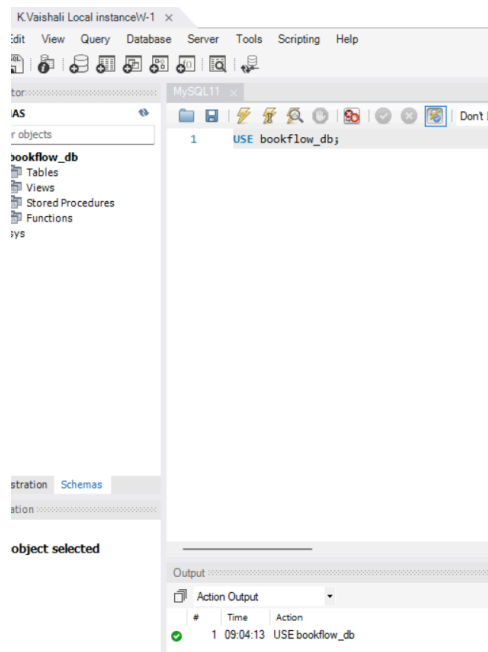


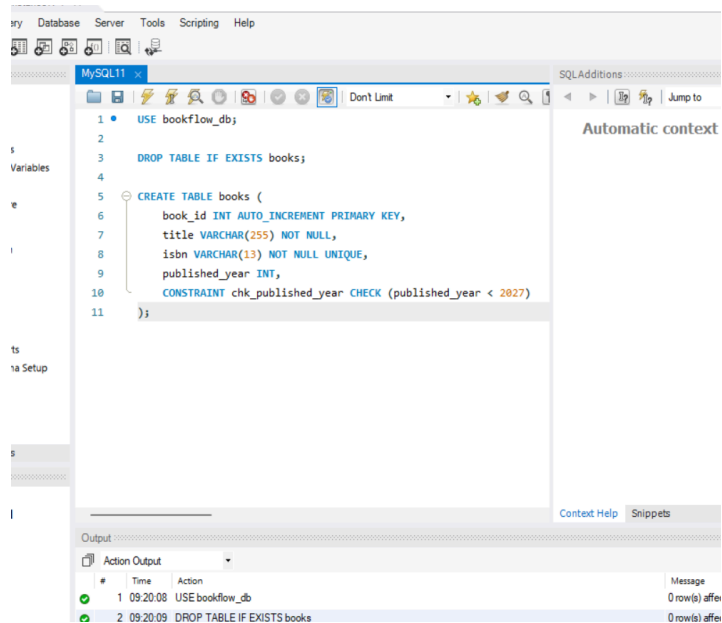
WEEK-1

Name-K.Vaishali
RollNo-2520030117
Section-7
subject-DBS&DBE

STEP-1 CREATE DATABASE



STEP-2 CREATE THE BOOKS TABLE



STEP-3 CREATE THE MEMBERS TABLE

KVaishali Local instanceW-1 x

File Edit View Query Database Server Tools Scripting Help

MySQL11*

```

1 DROP TABLE IF EXISTS members;
2
3 CREATE TABLE members (
4   member_id INT AUTO_INCREMENT PRIMARY KEY,
5   full_name VARCHAR(100) NOT NULL,
6   email VARCHAR(150) NOT NULL UNIQUE
7 );

```

Automatic context help is disabled. Use the toolbar to move the caret position or to toggle automatic context help.

Output

#	Time	Action	Message
1	09:20:08	USE bookflow_db	0 row(s) affected
2	09:20:09	DROP TABLE IF EXISTS books	0 row(s) affected
3	09:20:09	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, published_year INT(4) NOT NULL)	0 row(s) affected
4	09:21:26	DROP TABLE IF EXISTS members	0 row(s) affected, 1 warning(s):
5	09:21:26	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE)	0 row(s) affected

STEP-4 INSPECT THE TABLE STRUCTURES

KVaishali Local instanceW-1 x

Edit View Query Database Server Tools Scripting Help

MySQL11*

```

1 INSERT INTO books (title, isbn, published_year)
2 VALUES
3 ('Animal Farm', '9580061124153', 1992),
4 ('The Outsiders', '7013235088429', 2007),
5 ('The Pearl', '9780732211923', 2013);

```

Automatic context help is disabled. Use the toolbar to move the caret position or to toggle automatic context help.

Output

#	Time	Action	Message
1	09:20:08	USE bookflow_db	0 row(s) affected
2	09:20:09	DROP TABLE IF EXISTS books	0 row(s) affected
3	09:20:09	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, published_year INT(4) NOT NULL)	0 row(s) affected
4	09:21:26	DROP TABLE IF EXISTS members	0 row(s) affected, 1 warning(s): 1051 Unknown table 'bookflow_db.members'
5	09:21:26	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE)	0 row(s) affected
6	09:22:45	INSERT INTO books (title, isbn, published_year) VALUES ('Animal Farm', '9580061124153', 1992), ('The Outsiders', '7013235088429', 2007), ('The Pearl', '9780732211923', 2013);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

K.Vaishali Local instance\W-1

File Edit View Query Database Server Tools Scripting Help

Server Status
Client Connections
Users and Privileges
Status and System Variables
Data Export
Data Import/Restore

Server Startup / Shutdown
Server Logs
Options File

Performance Dashboard
Performance Reports
Performance Schema Setup

Navigation Schemas

Object selected

MySQL11

1 INSERT INTO members (full_name, email)

2 VALUES

3 ('Vaishali Lucky', 'vaishali.lucky@example.com'),

4 ('Sai Kruthi', 'sai.kruthi@example.com'),

5 ('Virat Kohli', 'virat.kohli@example.com');

SQLAdditions

Automatic context help is disabled. Use the toolbar to n
caret position or to toggle autom

Output

Action Output

#	Time	Action	Message
✓ 2	09:20:09	DROP TABLE IF EXISTS books	0 row(s) affected
✓ 3	09:20:09	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT N...	0 row(s) affected
⚠ 4	09:21:26	DROP TABLE IF EXISTS members	0 row(s) affected, 1 warning(s): 1051 Unknown table 'bookflow_db.members'
✓ 5	09:21:26	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(1...	0 row(s) affected
✓ 6	09:22:45	INSERT INTO books (title, isbn, published_year) VALUES ('Animal Farm', '9580061124153', 1992), ('The O...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 7	09:23:48	INSERT INTO members (full_name, email) VALUES ('Vaishali Lucky', 'vaishali.lucky@example.com'), ('Sai ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Info Session

K.Vaishali Local instanceW-1 x

Edit View Query Database Server Tools Scripting Help

MySQL11* x Administration - Data Import/Res...

1 SELECT * FROM books;

Result Grid

book_id	title	isbn	published_year
1	Animal Farm	9580061124153	1992
2	The Outsiders	7013235088429	2007
3	The Pearl	9780732211923	2013
NULL	NULL	NULL	NULL

Administration Schemas

K.Vaishali Local instanceW-1 x

Edit View Query Database Server Tools Scripting Help

MySQL11* x

1 SELECT * FROM books;
2 SELECT * FROM members;

Automatic context help is disabled. Use the toolbar to move the caret position or to toggle automatic context help.

Result Grid

member_id	full_name	email
1	Vaishali Lucky	vaishali.lucky@example.com
2	Sai Kruthi	sai.kruthi@example.com
3	Virat Kohli	virat.kohli@example.com
NULL	NULL	NULL

books 1 members 2 x

Output

#	Time	Action	Message
4	09:21:26	DROP TABLE IF EXISTS members	0 row(s) affected. 1 warning(s): 1051 Unknown table 'bookflow_db.members'
5	09:21:26	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100))	0 row(s) affected
6	09:22:45	INSERT INTO books (title, isbn, published_year) VALUES ('Animal Farm', '9580061124153', 1992), ('The Outsiders', '7013235088429', 2007), ('The Pearl', '9780732211923', 2013)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	09:23:48	INSERT INTO members (full_name, email) VALUES ('Vaishali Lucky', 'vaishali.lucky@example.com'), ('Sai Kruthi', 'sai.kruthi@example.com'), ('Virat Kohli', 'virat.kohli@example.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	09:25:06	SELECT * FROM books	3 row(s) returned
9	09:25:06	SELECT * FROM members	3 row(s) returned

STEP-8 CONSTRAINT TESTS

TEST-1 DUPLICATE ISBN

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL statement:

```
1 • INSERT INTO books (title, isbn, published_year)
2 VALUES ('Fake Copy', '9580061124153', 2000);
```

The Output tab displays the results of the query execution:

#	Time	Action	Message
6	09:22:45	INSERT INTO books (title, isbn, published_year) VALUES ('Animal Farm', '9580061124153', 1992);	(The ... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
7	09:23:48	INSERT INTO members (full_name, email) VALUES ('Vaishali Lucky', 'vaishali.lucky@example.com');	(Sai... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	09:25:06	SELECT * FROM books	3 row(s) returned
9	09:25:06	SELECT * FROM members	3 row(s) returned
10	09:28:09	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000);	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
11	09:28:15	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000);	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'

TEST-2 NULL TITLE

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL statement:

```
1 • INSERT INTO books (title, isbn, published_year)
2 VALUES (NULL, '1999999999', 2010);
```

The Output tab displays the results of the query execution:

#	Time	Action	Message
7	09:23:48	INSERT INTO members (full_name, email) VALUES ('Vaishali Lucky', 'vaishali.lucky@example.com');	(Sai... 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	09:25:06	SELECT * FROM books	3 row(s) returned
9	09:25:06	SELECT * FROM members	3 row(s) returned
10	09:28:09	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000);	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
11	09:28:15	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000);	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
12	09:29:59	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '1999999999', 2010);	Error Code: 1048. Column 'title' cannot be null

TEST-3 FUTURE PUBLICATION YEAR(CHECK VOILATION) && DUPLICATE EMAIL(UNIQUE VOILATION)

The screenshot shows the MySQL Workbench interface. The left sidebar contains a tree view with categories like 'Server Status', 'Client Connections', 'Users and Privileges', 'Status and System Variables', 'Data Export', 'Data Import/Restore', 'Server Tools', 'Server Logs', 'Options File', 'Performance Reports', and 'Performance Schema Setup'. The main window displays a SQL query in the 'SQL Editor' tab:

```
1 • INSERT INTO books (title, isbn, published_year)
2 VALUES ('Time Traveler', '8888888888', 2030);
```

The 'SQL Additions' tab on the right shows a message: 'Automatic context help is disabled. Use the toolbar icon to toggle automatic context help.' Below the editor, the 'Output' tab is active, showing a table with columns '#', 'Time', 'Action', and 'Message'.

#	Time	Action	Message
8	09:25:06	SELECT * FROM books	3 row(s) returned
9	09:25:06	SELECT * FROM members	3 row(s) returned
10	09:28:09	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
11	09:28:15	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
12	09:29:59	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '1999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
13	09:31:10	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.

Test-4 (duplicate email)

Step 1 — insert Rohit (should succeed):

K.Vaishali Local instance\W-1 x

File Edit View Query Database Server Tools Scripting Help

MySQL11 x

Don't Limit

```
1 INSERT INTO members (full_name, email)
2 VALUES ('Rohit Sharma', 'rohit.sharma@example.com');
```

Automatic context help is disabled. Use the toolbar to manually move the caret position or to toggle automatic context help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
9	09:25:06	SELECT * FROM members	3 row(s) returned
10	09:28:09	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
11	09:28:15	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
12	09:29:59	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '1999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
13	09:31:10	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.
14	09:31:42	INSERT INTO members (full_name, email) VALUES ('Rohit Sharma', 'rohit.sharma@example.com')	1 row(s) affected

Info Session

Step 2 — run the exact same query again (should fail):

K.Vaishali Local instance\W-1 x

File Edit View Query Database Server Tools Scripting Help

MySQL11 x

Don't Limit

```
1 INSERT INTO members (full_name, email)
2 VALUES ('Rohit Sharma', 'rohit.sharma@example.com');
```

Automatic context help is disabled. Use the toolbar to manually move the caret position or to toggle automatic context help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
10	09:28:09	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
11	09:28:15	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9580061124153', 2000)	Error Code: 1062. Duplicate entry '9580061124153' for key 'books.isbn'
12	09:29:59	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '1999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
13	09:31:10	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.
14	09:31:42	INSERT INTO members (full_name, email) VALUES ('Rohit Sharma', 'rohit.sharma@example.com')	1 row(s) affected
15	09:32:39	INSERT INTO members (full_name, email) VALUES ('Rohit Sharma', 'rohit.sharma@example.com')	Error Code: 1062. Duplicate entry 'rohit.sharma@example.com' for key 'members.email'

Info Session